RESEARCH WORKING PAPER / STP-008

The Archive Is Not the Population

Selection-Aware Event Memory for Industrial Edge Intelligence

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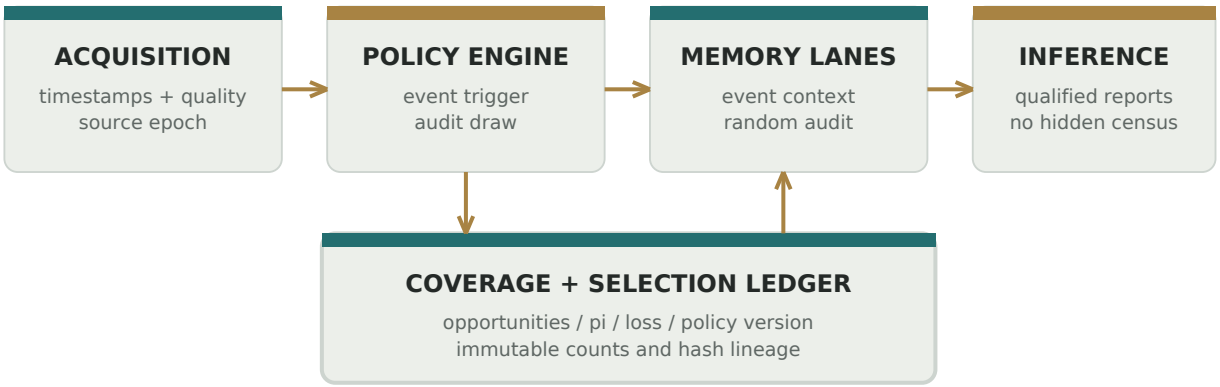
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Prepared: 2026-11-11 Version: 1.0 Status: WORKING PAPER

DOI: Not assigned (GitHub edition)

> CENTRAL THESIS | An event archive is a sample chosen by a capture policy. Unless that policy and its observation gaps remain visible, an apparently precise history can produce systematically misleading statistical claims.

Two capture lanes share one denominator and one audit contract.



Research synthesis and falsifiable architecture proposal. The diagram is conceptual; it does not depict a deployed SynapticSteel memory-selection system.

Abstract

An event-triggered industrial archive is often treated as the history of a machine, although it is actually the output of a selection policy. Thresholds, model alerts, buffer limits, sensor outages, operator actions, and retention rules determine which intervals survive. The resulting collection may be useful for forensic review yet unsuitable for estimating event prevalence, calibrating false-alarm rates, or comparing detectors: its inclusion mechanism is coupled to the behavior being inferred. This paper develops a Selection-Aware Memory Contract (SAMC) that treats capture eligibility, inclusion probability, acquisition coverage, retention transformations, and evidence provenance as separate first-class records. It combines deterministic event preservation with a bounded, randomized background audit lane and an explicit opportunity/coverage ledger. A worked numerical example shows how an archive can exaggerate apparent anomaly prevalence even when no data are fabricated; probability-weighted estimation is possible only under stated sampling and observability assumptions. The proposal synthesizes classical survey sampling and missing-data theory with modern anomaly-benchmark criticism, industrial data practice, and provenance standards. No empirical performance improvement or deployed SAMC capability is claimed. A falsifiable replay-and-field evaluation plan specifies when the architecture would justify further development.

Keywords: industrial edge intelligence; event memory; selection bias; informative missingness; sampling provenance; industrial anomaly detection; coverage ledger; Horvitz-Thompson estimation; reservoir sampling; bounded retention; evidence governance; time-series evaluation.

Suggested citation: Yarian, M. (2026). The Archive Is Not the Population: Selection-Aware Event Memory for Industrial Edge Intelligence. SynapticSteel Technical Papers, STP-008, version 1.0. Yarian Works, LLC. GitHub edition; no DOI assigned.

> Research status: this paper presents an architectural synthesis, an exact illustrative arithmetic example, and a proposed evaluation. It reports no new field experiment, no measured detection lift, and no validated estimator on SynapticSteel equipment.

Contribution and scope

The proposed contribution is not a new sampling estimator or a new anomaly detector. Classical probability sampling, missing-data theory, and provenance representation already exist [1-4]. The proposed integration is an evidence contract for resource-constrained industrial memory: an explicit capture denominator, logged inclusion decisions, distinct acquisition versus retention gaps, and a claim gate that prevents case archives from being mistaken for representative operating history.

A machine may have multiple defensible histories: a forensic history optimized for retaining unusual intervals, a statistical history optimized for estimating population properties, and a provenance history documenting which records could have existed. Those histories answer different questions. A selection-aware design should preserve the distinction rather than compress them into one deceptively authoritative event timeline.

1. The quiet sampling mechanism

An industrial edge device cannot always retain every high-rate vibration waveform, audio frame, thermal image, inference trace, and operator note. It therefore records summaries periodically, captures detailed context around alarms, rotates logs, and archives or suppresses cases according to policy. Each decision is individually reasonable. Together, they construct a selection mechanism: the retained archive overrepresents conditions that the capture policy considered interesting, affordable, or reachable.

This is not necessarily a software defect. Event-focused archives are efficient for forensic investigation. The defect appears when a later analyst treats the selected collection as if it were an unbiased sample of all machine operation. A model trained only on retained incidents may learn the detector that selected those incidents. A false-alarm estimate computed only from reviewed cases may omit ordinary hours and unreviewed events. A report that counts episodes without the total observation opportunities cannot report an operating prevalence.

The concern is more specific than missing sensor data. A sensor can be healthy and every retained record can be accurate while the archive remains systematically unrepresentative. Conversely, a sensor outage can make the population itself partially unobserved, so no statistical reweighting of retained records can recover the unmeasured physical truth without additional assumptions. Rubin’s missingness framework formalizes why the observation mechanism matters [1].

A motivating example

Consider a pump that runs for ten thousand one-second windows. An anomaly trigger saves every flagged window, while a background audit saves a small random fraction of unflagged windows. The archive is excellent for reviewing trigger episodes. But its proportion of anomalous windows is a mixture of the machine’s behavior and the capture rule. If the threshold changes mid-shift, the archive changes even when the physical machine does not. The inverse problem is not solved by making the archive larger; it is solved, where identifiable, by preserving how the sample was selected.

2. Prior work and competing designs

Missingness and unequal-probability sampling

Rubin distinguishes cases in which missingness can be ignored for a particular inference from cases in which it cannot [1]. Horvitz and Thompson provide a design-based estimator for finite populations when units have known, nonzero inclusion probabilities [2]. Vitter’s reservoir sampling provides a practical one-pass method for choosing records from a stream of unknown length [3]. These results are complementary: the first asks whether absence is informative, the second specifies when unequal sampling can be corrected, and the third offers a bounded random selection mechanism.

None of these results grants automatic validity to an event archive. If selection depends on unrecorded physical severity, if some intervals had zero chance of retention, or if acquisition failed before any capture decision, inclusion probabilities for the intended population may be unknown or zero. Weighting a biased sample with guessed probabilities can make a result look scientific while increasing error.

Provenance and anomaly-benchmark skepticism

W3C PROV-O supplies a general language for entities, activities, agents, and derivations [4]. It can describe a sensor observation, a trigger policy, and a retention transformation, but it does not prescribe a statistically adequate inclusion ledger. Separately, Kim et al. show how common point-adjustment evaluation can inflate time-series anomaly scores [5], while Wu and Keogh identify flaws in benchmark composition and labeling [6]. NoBOOM offers real industrial process datasets and explicitly examines benchmark bias [7]. These studies motivate evaluation discipline; none validates the architecture proposed here.

Recent missing-data benchmarking also warns against judging algorithms only on conveniently simulated random deletion: Toye, Celik, and Kleinberg find meaningful differences when missingness mechanisms change [8]. The application domain differs from industrial machinery, so its empirical numbers are not transferable. The transferable lesson is methodological: evaluate the observation mechanism that actually exists.

Design alternatives and trade-offs

Approach	Strength	Failure mode for inference
Retain everything	Potentially strongest replay and audit fidelity	May exceed bandwidth, storage, privacy or compute limits; acquisition gaps still exist
Event-only capture	Efficient forensic context around flagged episodes	No denominator; trigger policy controls apparent prevalence
Uniform reservoir	Known design under appropriate stream assumptions	Rare failures may be missed; contextual bursts are fragmented
Event + random audit	Keeps forensic detail and a probabilistic baseline	Requires explicit opportunity counts, sampling logs and coverage checks
Model-based imputation	May recover useful forecasts under validated assumptions	Can conceal informative missingness and fabricate apparent continuity

A hybrid event-plus-audit design is a proposal, not a universal optimum. For small low-rate systems, complete retention may be simpler and superior. For highly safety-critical systems, regulatory and forensic requirements may require lossless logging of specified channels regardless of the statistical audit lane.

3. Formal problem: the archive as a selected sample

Partition a declared operating horizon into eligible windows i = 1, ..., N. Let Y_i be a property of the physical interval that an investigator eventually wants to estimate, such as a verified anomaly label. Let X_i denote physical state, Z_i a pre-retention trigger proxy, A_i an indicator that acquisition actually observed the interval, S_i the policy-driven retention decision given acquisition, and R_i the indicator that the selected record survived storage and export. The final accessible indicator is O_i = A_i S_i R_i. These variables describe different mechanisms and must not be collapsed into one “missing” flag.

```
O_i = A_i * S_i * R_i ; pi_i = P(S_i = 1 | A_i = 1, logged design state)
```

A retained-only statistic generally estimates E[Y | O = 1], not E[Y] over eligible operating windows. If a trigger proxy Z is correlated with Y, then P(S = 1 | Y) can differ across physical states even when the trigger is perfectly deterministic. If A_i = 0, Y_i may be unobserved regardless of the retention policy. If R_i is driven by disk pressure that coincides with high load, storage loss can also be informative. Each layer needs separate accounting.

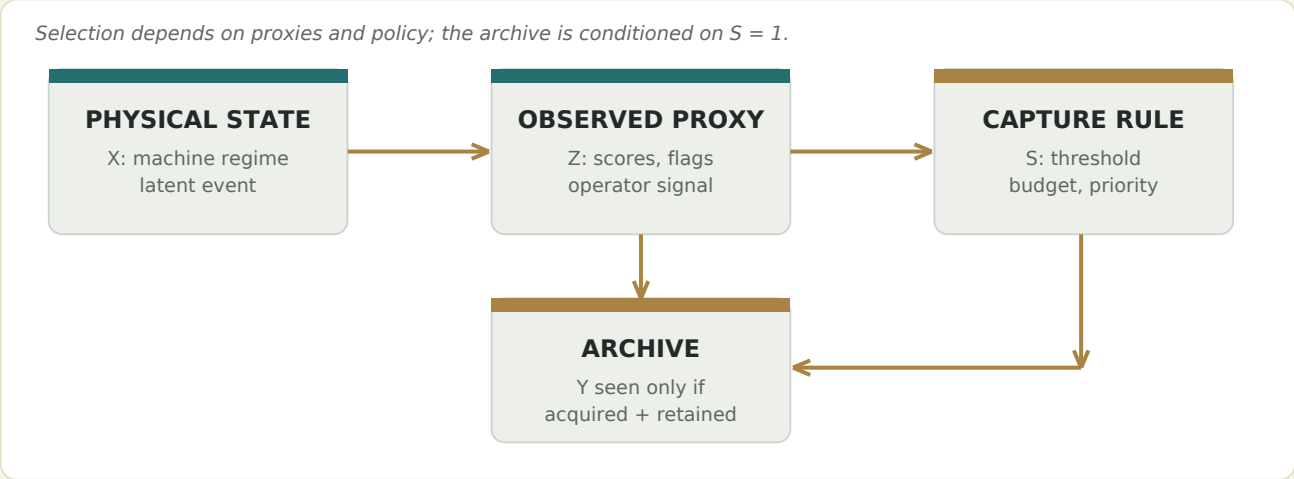


Figure 1. Conceptual selection graph. Physical state influences the trigger proxy; the policy uses that proxy to decide what is stored. The archive is therefore conditioned on the policy. A separate acquisition-loss path, omitted for readability, can make some physical intervals unobservable.

Identifiability boundary

A design-based estimator can be defended for an acquired population if each eligible acquired window has known positive inclusion probability, the random draw is implemented as specified, and downstream loss is either absent or explicitly modeled. To generalize to all physical operating time, acquisition coverage must also be adequate or independently modeled. A device cannot estimate what occurred during an unobserved interval merely by logging that the interval was absent.

A further subtlety is the unit of analysis. If event captures overlap, a window can be selected by several triggers. Its inclusion probability is the probability of being retained at least once, not the probability of any single trigger firing. If sampling is done by bursts, neighboring windows are dependent. Treating them as independent one-second draws understates uncertainty.

4. A numerical counterexample, not an experiment

The following values are invented solely to make the selection effect auditable. Assume an oracle later establishes the true anomaly labels for 10,000 windows. A trigger proxy flags 500 windows, of which 150 are anomalous. The other 9,500 windows contain 50 anomalies. The policy retains all 500 flagged windows and independently audits 10% of unflagged windows. In this particular illustrative draw, the audit includes five anomalous and 945 normal windows. This realized sample is possible, but it is not a measurement from any TelePod or published dataset.

Stratum	Population windows	True anomalies	Retained windows	Retained anomalies
Flagged, pi = 1.0	500	150	500	150
Unflagged, pi = 0.1	9,500	50	950	5

Stratum	Population windows	True anomalies	Retained windows	Retained anomalies
Total	10,000	200	1,450	155

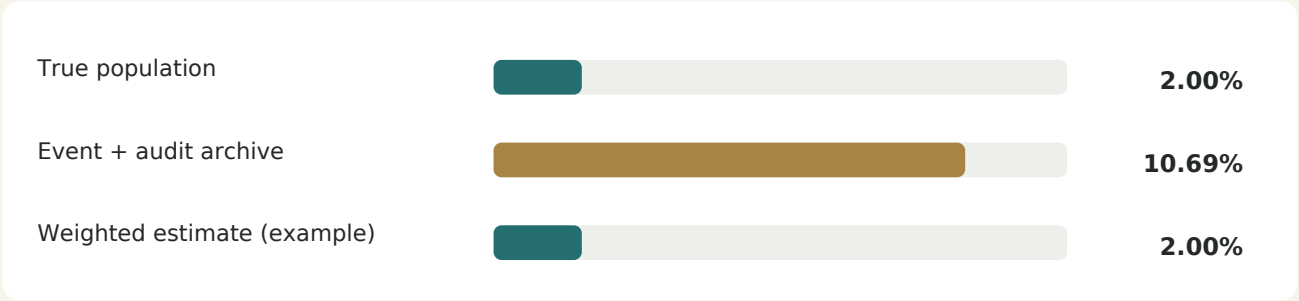


Figure 2. Exact arithmetic for one constructed realization. The retained archive shows 155 / 1,450 = 10.69% anomalies, although the population contains 200 / 10,000 = 2.00%. This is selection distortion, not model performance.

```
Naive archive fraction = 155 / 1450 = 10.69% ; population fraction = 200 / 10000 = 2.00%
```

Because the policy uses a known 10% independent random audit for unflagged windows, a Horvitz-Thompson-style total estimator can use weight 1 for flagged records and weight 10 for audited unflagged records [2]. For this particular draw, the estimated anomaly count is 150 + 5 / 0.1 = 200 and the estimated eligible-window count is 500 + 950 / 0.1 = 10,000. The ratio is 2.00%. That coincidence is not guaranteed in other draws; the estimator has sampling uncertainty and its ratio form need not be exactly unbiased.

```
T_hat(Y) = sum_(i in retained) Y_i / pi_i ; p_hat = T_hat(Y) / T_hat(1)
```

This correction is legitimate only because the inclusion rule was defined and logged on a variable available before retention. In real deployments, an anomaly label is not an oracle available to the capture system. The sampling probability must be derived from the actual randomized policy and eligibility strata, not estimated from the post-hoc labels of the retained archive. If unflagged anomalies are never sampled, the population anomaly rate is not identifiable from the archive alone.

A low audit fraction also has a practical failure mode: in rare-event strata, a sample may contain zero positive labels. An unbiased total estimator in expectation does not imply a useful interval estimate from one short deployment. Variance, temporal clustering, finite-horizon coverage and zero-inclusion strata must be reported; they cannot be repaired by an attractive point estimate.

5. Selection-Aware Memory Contract (SAMC)

SAMC is a proposed minimal contract among acquisition, capture policy, retention, and reporting. Its central rule is that every statistical claim must name its target population and show the observation/selection path that makes that population estimable. Forensic event detail may remain high priority, but it must not silently become the denominator for model performance.

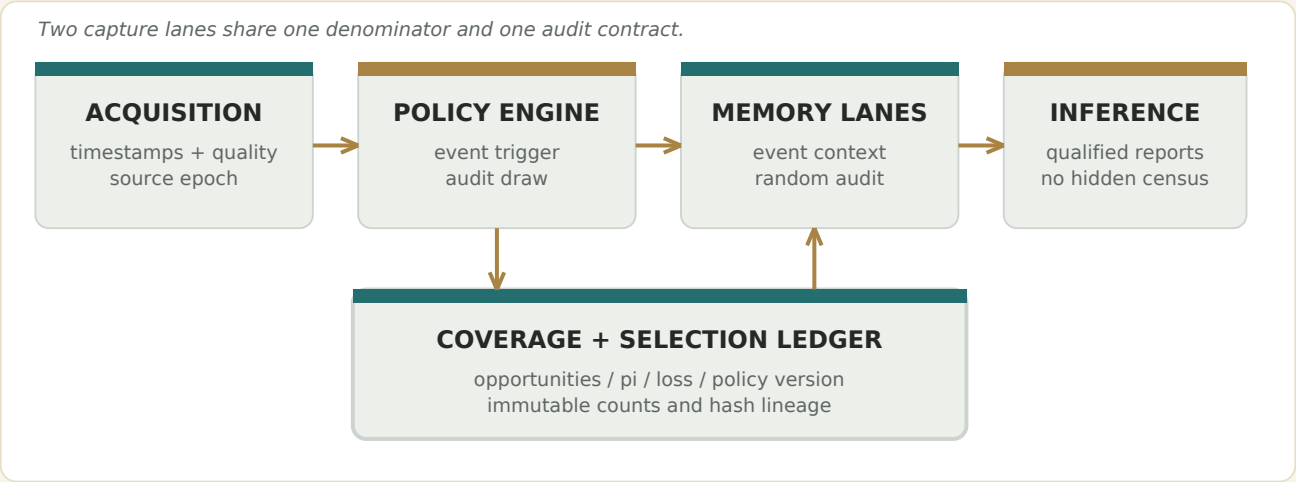


Figure 3. Proposed architecture. The event lane preserves rich incident context; a randomized audit lane samples ordinary operation. A separate coverage and selection ledger records eligibility, inclusion rules, policy revisions, and loss. Reporting admits only claims whose denominator and uncertainty are supportable.

Five proposed invariants

I1 - Declared opportunity unit. Each capture policy specifies a measurable eligible unit: a second, a waveform block, a machine cycle, or an operating-regime interval. Counts across different units are never silently pooled. I2 - Inclusion semantics. Each sampled unit records its policy version, trigger features, randomization stratum and inclusion probability or a reason that probability is unknown. I3 - Gap separation. Acquisition absence, deliberate nonselection, postselection storage loss, and privacy-driven redaction are distinct states. I4 - Provenance integrity. A report can trace each selected record through transformations and archive generations; checksums prove byte preservation, not representativeness. I5 - Claim gating. A report must label estimates as population-representative, acquired-population-only, selected-case descriptive, or non-estimable.

Capture lanes and failure behavior

The event lane retains bounded pre/post context for triggers, with deduplication and explicit overlapping-window treatment. The audit lane makes a reproducible random draw from the eligible unflagged stream, with a documented seed or auditable pseudorandom source, inclusion rate and sampling unit. The coverage ledger increments opportunity counters before any retention decision, records observed acquisition gaps, and periodically emits a compact signed or hash-linked checkpoint. If ledger writes fail, the system must not silently continue issuing population-level statistics.

An explicit emergency policy may suspend random audit to protect the device. That decision must itself be logged as a change in the estimand and a coverage break. Safety and resource limits outrank the statistical elegance of the audit. The contract is designed to fail closed for unsupported claims, not to keep every capture lane running regardless of consequences.

Minimum proposed record

```
{window_id, source_epoch, regime_id, acquisition_state, trigger_policy_id,
```

```
selection_stratum, inclusion_probability, capture_reason, retention_state,
```

```
transform_hash, time_support, coverage_checkpoint_id, claim_scope}
```

This is a logical schema, not a mandate to write one JSON object per raw sample. A low-resource implementation can aggregate eligibility counters by regime and policy revision, log per-selected-window decisions, and emit compact checkpoints. A cryptographic digest can detect unexpected changes to retained bytes, but without an external immutable anchor it does not prove that omitted events never existed.

6. Statistical use and explicit refusal rules

Under an independently randomized audit of acquired eligible units with known nonzero probabilities, design-based totals or stratified estimators can be considered. The audit should record enough structure to estimate uncertainty under the actual design. If bursts or machine cycles are the sampling units, variance estimation must respect clusters. If deterministic event capture overlaps randomized audit, the inclusion probability must reflect the union of selection paths, or the analysis must partition the frame into disjoint strata before sampling.

When acquisition is incomplete, the target population should be restricted explicitly to the acquired frame unless there is a defensible independent model for missing physical intervals. Missing-not-at-random mechanisms cannot generally be resolved by reweighting observed records alone [1,8]. A report may still present event counts and forensic narratives, but it should not convert them into machine-wide prevalence or calibrated probabilities.

Requested claim	Minimum evidence	Safe response if absent
What happened in this retained event?	Event payload + provenance + time/quality support	Descriptive account with uncertainty
How often did anomalies occur?	Population frame, coverage, known pi, labels, variance	Non-estimable; report selected-case counts
Did model A beat baseline B?	Common independent labeled horizon + fixed scoring	No performance conclusion
Was a quiet hour healthy?	Acquisition coverage + validated sensitivity	No anomaly observed is not proof of health
Was archive restoration complete?	Manifest hashes + full restore reconciliation	Integrity claim only for tested archive scope

For evaluation, temporal leakage is a particular danger. A model must not train on the same alert-triggered capture that defines its evaluation set. Split by physical run, machine and time before model selection. Do not inflate scores through point-adjustment procedures that reward any single hit in a long anomalous segment [5]. Use simple statistical baselines and report time-to-detection, false alarms per operating hour, missed events and confidence intervals on an independently labeled horizon.

7. Falsifiable evaluation program

Hypotheses and baselines

H1: under a fixed storage budget, event-plus-audit memory yields lower error in estimated anomaly prevalence than event-only capture, provided acquisition coverage and design probabilities are known. H2: selection-aware evaluation changes at least some detector rankings relative to case-only scoring when capture is correlated with the detector. H3: adding a coverage ledger and audit lane can remain within a predeclared edge resource envelope without degrading acquisition availability. H4: deliberate acquisition outages cause SAMC to refuse unsupported whole-horizon prevalence estimates rather than fabricate a complete history.

Compare four designs on identical replay: B0 full-fidelity reference retained off-device for experimental ground truth; B1 event-only capture; B2 uniform fixed-budget reservoir; B3 event-plus-random-audit with selection ledger. B0 is a laboratory reference, not a deployable assumption. Hold total storage, CPU time and acquisition cadence constant for B1-B3 where feasible. Compare the same detectors, including a simple threshold/EWMA or change-point baseline and a more complex temporal model, without allowing capture policy to use the evaluation labels.

Test matrix

Trial	Manipulation	Primary measurement
A - clean stream	Known event rate, stable acquisition	Prevalence error, variance calibration
B - policy shift	Trigger threshold changes mid-run	Bias before/after policy revision
C - clustered events	Bursts, overlap, regime transitions	Inclusion accounting and interval coverage
D - blind outage	Drop sensor upstream of policy	Correct refusal of whole-horizon inference
E - resource stress	Near-full disk, restart, rotation	Loss ledger accuracy, CPU/RAM/storage
F - model leakage	Evaluate on selected cases vs independent labels	Rank stability, false alarms/hour

Primary metrics: absolute prevalence error against B0; 90% or 95% confidence-interval coverage over repeated independent runs; event-context recall under equal bytes; CPU peak and resident-memory overhead; dropped acquisition windows; ledger reconciliation rate; unsupported-claim refusal precision; and detection performance on a separately labeled, unselected test horizon. Pre-register acceptable resource budgets and the number of repetitions before inspecting outcomes. An engineering failure occurs if the audit lane worsens physical acquisition stability or if inclusion probabilities cannot be reconstructed.

Use both controlled synthetic streams and at least one publicly described industrial benchmark such as NoBOOM, subject to its licensing and the suitability of available raw time series [7]. Do not treat a benchmark’s anomaly labels as exhaustive machine truth without checking its annotation protocol. A real dual-device trial should be strictly shadow/read-only at first and should include operator-reviewed physical perturbations only under separate authorization. A reproducible implementation and measured outcomes are prerequisites for a stronger empirical paper.

8. Counterarguments, costs and failure modes

Counterargument 1: “We only need incident investigation, not population statistics.” That is valid. An event-only archive can be the right design when all downstream claims remain case-descriptive. SAMC becomes valuable when the same archive is later used for training, drift estimates, prevalence, or comparative model scoring. The proposed response is not to force sampling everywhere, but to encode a clear boundary around unsupported uses.

Counterargument 2: “We can retain every summary, so no audit lane is needed.” For low-rate summaries this may be correct. Yet summaries can suppress subsecond anomalies and may themselves be produced by an adaptive algorithm. Full retention of a derived stream does not imply full observation of the physical process. A coverage ledger is still useful, but the random audit lane may be unnecessary if the declared target population is genuinely captured completely.

Counterargument 3: “Known inclusion probabilities solve the bias.” Only under the relevant design assumptions. Unknown physical events, zero-probability windows, policy bugs, sensor outages, burst dependence and model-dependent triggers can invalidate an apparently elegant estimator. Design-based correction is a narrow instrument, not a universal cure. In particular, sampling design probabilities do not substitute for ground-truth labels or calibrated detector sensitivity.

Counterargument 4: “More logging makes the system less safe.” This is a real risk in OT. NIST’s final 2023 OT guidance emphasizes availability, performance, and safety constraints [9]; the September 2026 Revision 4 is an initial public draft, not final guidance [10]. A practical design must bound writes, defer or disable audit under resource stress, preserve primary acquisition, and report any resulting loss of inferential coverage. A tiny opportunity counter may be preferable to detailed per-window audit events on constrained hardware.

Additional threats include pseudorandom generator resets that repeat audit selections; policy revisions that are not versioned; log rotation that discards the only denominator; duplicate episodes counted as independent events; user-requested redaction that changes the analysis frame; and malicious manipulation of selection logs. Hashes and provenance reduce ambiguity but do not prove physical completeness. Privacy may require retaining only aggregated coverage rather than raw content, with explicit limits on what can then be inferred.

9. Experimental scope and implementation boundary

This paper is a research proposal and evaluation protocol. Private operational checkpoints are excluded from this public edition. No implementation, deployment, field validation, or measured performance benefit is established by the proposal. A prototype should first be evaluated with recorded or simulated inputs, explicit baselines, failure cases, and independently measured outcomes.

10. Limitations, conclusions and research classification

The contract cannot create evidence from unobserved physical time. It adds state, storage writes and analysis complexity, and its value depends on the intended downstream claim. A random audit lane may miss rare events at feasible sampling rates; event preservation can distort the audit if inclusion accounting is wrong. The exact audit probability, window definition, time support and resource threshold must be chosen for each device and operating regime. None is specified as a universally safe default.

The defensible conclusion is narrower than “sampling fixes anomaly detection.” Event-triggered memory is a designed sample. Forensic usefulness and statistical representativeness are distinct properties. When retention policy, coverage and acquisition loss are explicit, an analyst can sometimes estimate population properties under testable design assumptions; when they are not, the honest result is a bounded description of selected cases. This is a stronger foundation for industrial machine memory than silently promoting the archive into a census.

> Research classification: SSRN working-paper candidate after author/editorial review. An arXiv submission or conference/journal claim of methodological novelty would require a reproducible implementation, registered experiments, independent labels, repeated trials, uncertainty calibration and a compelling comparison against simple full-log and reservoir baselines. No peer review or venue acceptance is implied.

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Related SynapticSteel Technical Papers

Related series entries (local identifiers are not DOIs; publication states vary): STP-001, Failure Metabolism; STP-003, The Timestamp Is Not the Event; STP-004, Silence Is a Signal, But Not a Measurement; STP-005, Trust Is Not Agreement; STP-006, The Probe Is Part of the Process; STP-007, A Reconnection Is Not a Recovery. The present contribution addresses archive-selection mechanisms, not sensor credibility, timing, observation intervention, or reacquisition.

Publication and provenance note

Author-released working paper. GitHub edition published 11 November 2026. Not peer reviewed; no empirical performance improvement or production deployment is established. No DOI is assigned to this GitHub edition. Cite its versioned repository record. Copyright Mark Yarian. All rights reserved; public reading and citation permitted.